

BIG PINEY-MARBLETON, WY, USA

WMO: 726710

Lat: 42.584N

Lon: 110.108W

Elev: 6970

StdP: 11.35

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24164

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.3	-10.8	-20.6	2.3	-15.5	-15.4	3.1	-9.8	26.4	28.7	22.9	26.4	3.0	80	0.719

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.0	86.2	55.6	83.5	55.0	81.0	54.7	59.2	77.6	57.5	76.3	56.2	75.3	11.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.7	79.6	63.1	51.1	72.2	62.0	48.3	65.1	61.4	28.9	77.0	27.7	76.0	26.7	74.9	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.8	20.9	18.5	DB	-24.3	90.0	7.2	2.3	-29.5	91.7	-33.7	93.0	-37.7	94.2	-43.0	95.9	(4)
(5)			WB	-24.1	61.6	6.3	2.7	-28.7	63.6	-32.4	65.1	-36.0	66.6	-40.6	68.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.6	13.1	16.2	27.8	37.1	46.1	55.1	63.4	60.8	52.0	39.3	25.6	13.9
	DBStd	19.22	10.26	10.53	9.15	6.78	7.13	6.55	4.24	4.60	6.24	7.55	9.95	10.33
	HDD50	5602	1145	948	689	389	162	29	0	2	50	337	733	1118
	HDD65	10028	1610	1368	1154	837	587	298	79	143	389	797	1183	1583
	CDD50	1088	0	0	0	1	39	183	415	335	110	5	0	0
	CDD65	43	0	0	0	0	0	3	29	11	0	0	0	0
	CDH74	2893	0	0	0	0	27	420	1393	890	161	2	0	0
	CDH80	718	0	0	0	0	2	89	419	193	15	0	0	0
Wind	WSAvg	6.8	5.5	5.7	6.9	8.5	8.3	8.4	7.4	7.0	6.7	6.6	5.9	5.1
Precipitation	PrecAvg	8.8	0.5	0.5	0.5	0.6	1.3	1.0	0.7	0.9	0.9	0.8	0.4	0.5
	PrecMax	14.1	1.0	1.0	1.3	1.3	3.8	4.1	1.8	2.9	1.8	2.6	0.9	1.3
	PrecMin	4.4	0.2	0.1	0.1	0.1	0.1	0.1	0.2	0.1	0.1	0.0	0.0	0.1
	PrecStd	2.7	0.3	0.3	0.3	0.4	0.9	0.9	0.4	0.6	0.5	0.6	0.2	0.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.8	46.7	60.4	68.4	77.3	86.7	90.4	88.1	82.0	71.8	57.4	44.9
		MCWB	33.0	35.9	41.3	45.6	51.9	54.7	58.0	57.2	54.5	49.3	41.1	34.6
	2%	DB	37.1	39.7	53.3	63.4	73.0	82.3	87.7	84.3	78.4	65.5	52.0	38.0
		MCWB	30.5	32.7	38.5	43.4	49.9	54.1	56.3	54.4	51.7	45.9	38.8	31.3
	5%	DB	33.2	36.2	47.7	58.7	69.7	79.1	84.5	82.0	75.2	61.2	46.4	34.1
		MCWB	28.5	30.9	36.5	41.2	48.3	52.9	55.7	54.6	50.4	43.5	36.5	29.5
	10%	DB	29.5	32.5	43.0	53.9	64.4	75.2	81.9	79.6	71.8	56.1	41.6	30.1
		MCWB	26.0	28.4	34.3	39.6	46.2	51.6	55.4	53.7	49.5	41.6	33.6	26.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.8	36.6	42.6	47.1	54.8	57.8	62.3	61.1	57.5	50.1	42.9	35.5
		MCDB	40.2	44.3	55.5	66.2	74.2	80.8	82.9	80.4	75.1	66.9	55.0	42.8
	2%	WB	31.3	33.9	39.8	44.3	51.0	55.7	59.9	58.7	55.0	47.4	39.6	32.3
		MCDB	36.0	39.2	51.2	60.5	68.8	77.1	78.7	76.3	71.6	61.7	49.6	37.5
	5%	WB	28.5	31.2	37.2	42.2	48.9	54.3	58.3	57.1	52.8	45.0	37.0	29.8
		MCDB	32.7	35.4	46.7	56.5	65.9	74.6	77.2	74.6	68.8	58.0	45.5	33.7
	10%	WB	25.9	28.6	34.7	40.2	47.0	52.7	56.9	55.5	50.8	42.5	34.2	26.8
		MCDB	29.5	32.4	42.2	52.3	62.5	72.1	76.3	73.7	66.6	54.6	41.3	30.1
Mean Daily Temperature Range	5% DB	MDBR	25.8	25.1	24.6	27.2	29.0	33.6	36.0	36.1	34.1	29.5	26.8	25.1
		MCDBR	27.4	24.8	31.1	37.0	38.2	41.3	40.7	40.7	41.5	38.5	31.9	26.5
		MCWBR	21.7	18.4	18.4	19.6	18.3	18.4	17.1	17.9	19.7	21.4	20.4	20.4
	5% WB	MCDBR	26.3	23.4	28.4	33.0	34.1	37.5	34.6	34.0	34.3	34.2	30.0	25.2
		MCWBR	21.2	17.9	17.7	18.3	17.0	17.5	15.4	15.9	16.9	19.5	19.6	19.7
Clear-Sky Solar Irradiance	taub	0.227	0.237	0.253	0.285	0.302	0.304	0.329	0.324	0.294	0.264	0.238	0.224	
	taud	2.387	2.372	2.442	2.444	2.475	2.505	2.453	2.456	2.535	2.590	2.553	2.452	
	Ebn at Noon	299	313	319	313	308	306	297	296	300	300	295	292	
	Edh at Noon	24	29	31	34	34	33	35	33	29	23	21	21	
All-Sky Solar Radiation	RadAvg	670	953	1377	1738	1964	2344	2304	1993	1610	1086	713	541	
	RadStd	47	91	86	99	169	176	120	87	123	79	60	43	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page